



Department of Computer Science and Engineering

Academic Year 2023– 2024(Odd Semester)

Degree, Semester & Branch: B.E – I Semester Mech

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs. M.Dhivya, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Guess an Integer number in a range
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Role play
- **Justification:**
 - Role Play is the activity where the individual takes the role and act according to the given role.
 - **Guess an integer number in a range** is a kind of game where the students are involved to play the game as per the rule.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - Faculty explained the concept of the problem. Two students were asked to come in front of the classroom.
 - Student 1 was asked to fix a random number 'x' in the range of 1 to 50
 - Student 2 was asked to guess the number 'y'. If the guessed number 'y' is greater than the random number 'x' then student 1 says "Too high".
 - If the guessed number 'y' is less than the random number 'x', then student 1 says "Too low"
 - If the guessed number 'y' is equal to the random number 'x', then the student 1 says "You win"
 - To make the activity more interesting, number of guesses is fixed. If the student 2 fails to guess a number within the fixed number of guesses, the student 1 says "You lose"
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PSO1
CO1	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)



- **PO / PSO mapped:**

Innovative practice	Justification for correlation
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO9	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PSO1	Students will be able to provide sustainable solution to mechanical engineering problem using the various algorithmic tools.

- **Images / Screenshot of the practice:**



Fig 1: Students involvement in Number Guessing Game



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students said it is quiet easy to understand when compare with other illustrative programs
- Through this activity the students felt easy to write the program

- ❖ ***Benefit of the practice:***

- Students understanding of the problem is improved
- Students coding skill through this activity is better

- ❖ ***Challenges faced in implementation:***

- This activity involves only 2 members
- Unable to make all the students to involve in this activity

References:

- ❖ <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS8493-OS-Roleplay.pdf>
- ❖ https://aminoapps.com/c/attack-on-titan/page/blog/different-types-of-roleplay-lit-team/YoYg_nMTbux63E26M7z6xpRxkovvoBXPj